

ASSESSMENT-2

TOOL

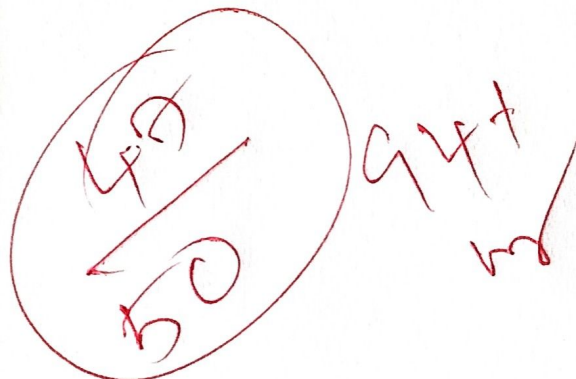
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course code :- MCA0401

course :- Deep learning



Introduction

Neural network is a machine learning model that learns pattern from data and predict output.

during training - model learns.

during testing - predicts result

$$\text{Training MSE} = 0.85$$

$$\text{Testing MSE} = 5.20$$

MSE (mean squared Errors)

$$\text{MSE} = \frac{1}{n} \sum (\text{Actual} - \text{predicted})^2$$

lower MSE - better predicted.

Higher MSE - large predicted Errors.

Given,

$$\text{Training MSE} = 0.85$$

$$\text{Testing MSE} = 5.20$$

$$5.20 > 0.85$$

the model has high testing error.

$\therefore$ It is overfitting.

Reasons for High Testing Error

- overfitting
- Small Training Dataset
- Complex neural network.
- noise in data.
- lack of Regularization.
- Improper data Splitting.

corrective measures

- Increase training data
- use regularization
- Early stopping
- Simplify the model
- Remove noise
- Tune hyperparameters.

Training error	Testing error	Interpretation
High	High	underfitting
low	High	overfitting
low	low	good model

This model is overfitting.

conclusion :-

Since the testing error is much larger than the training error, the model is suffering from overfitting.